

MEMORY

SINGLE PORT BRAM (16x2048)

//Verilog code [memory1.v]

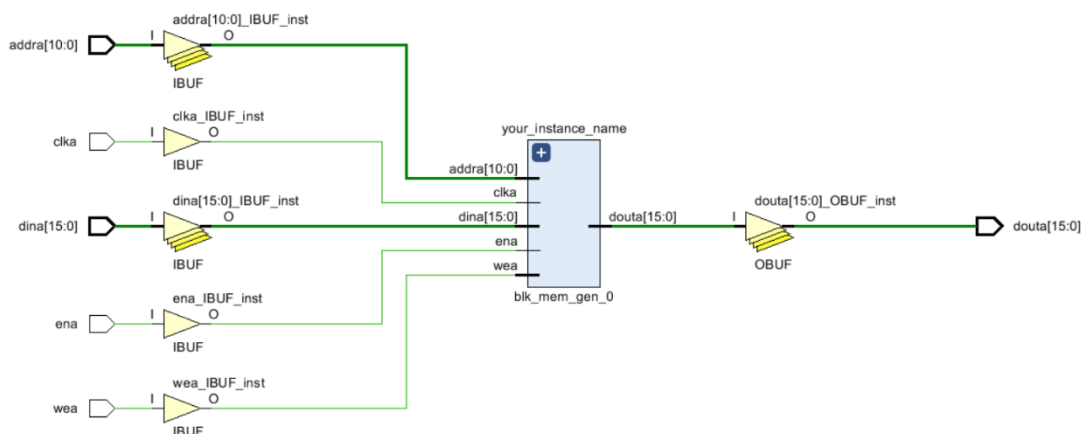
`timescale 1ns / 1ps

```
module memory1(
    input clka, ena, wea,
    input [10:0]addra,
    input [15:0]dina,
    output [15:0]douta);
    blk_mem_gen_0 your_instance_name (
        .clka(clka), // input wire clka
        .ena(1'b1), // input wire ena
        .wea(wea), // input wire [0 : 0] wea
        .addra(addra), // input wire [10 : 0] addra
        .dina(dina), // input wire [15 : 0] dina
        .douta(douta) // output wire [15 : 0] douta
    );
endmodule
```

2. Memory

Site Type	Used	Fixed	Available	Util%
Block RAM Tile	1	0	140	0.71
RAMB36/FIFO*	1	0	140	0.71
RAMB36E1 only	1			
RAMB18	0	0	280	0.00

RTL analysis schematic:



2 DUAL-PORT BRAM (32x16k)

//Verilog code [memory2.v]

`timescale 1ns / 1ps

```
module memory2( input clka, clkb, ena, enb, wea, web, input [13:0]addra, input [31:0]dina, output [31:0]douta,
input [13:0]addrb, input [31:0]dinb, output [31:0]doutb);
```

```
    blk_mem_gen_2 your_instance_name (
        .clka(clka),    // input wire clka
        .ena(ena),      // input wire ena
        .wea(wea),      // input wire [0 : 0] wea
        .addra(addra),  // input wire [13 : 0] addra
        .dina(dina),    // input wire [31 : 0] dina
        .douta(douta),  // output wire [31 : 0] douta
        .clkb(clkb),    // input wire clkb
        .enb(enb),      // input wire enb
        .web(web),      // input wire [0 : 0] web
        .addrb(addrb),  // input wire [13 : 0] addrb
        .dinb(dinb),    // input wire [31 : 0] dinb
        .doutb(doutb)   // output wire [31 : 0] doutb
    ); endmodule
```

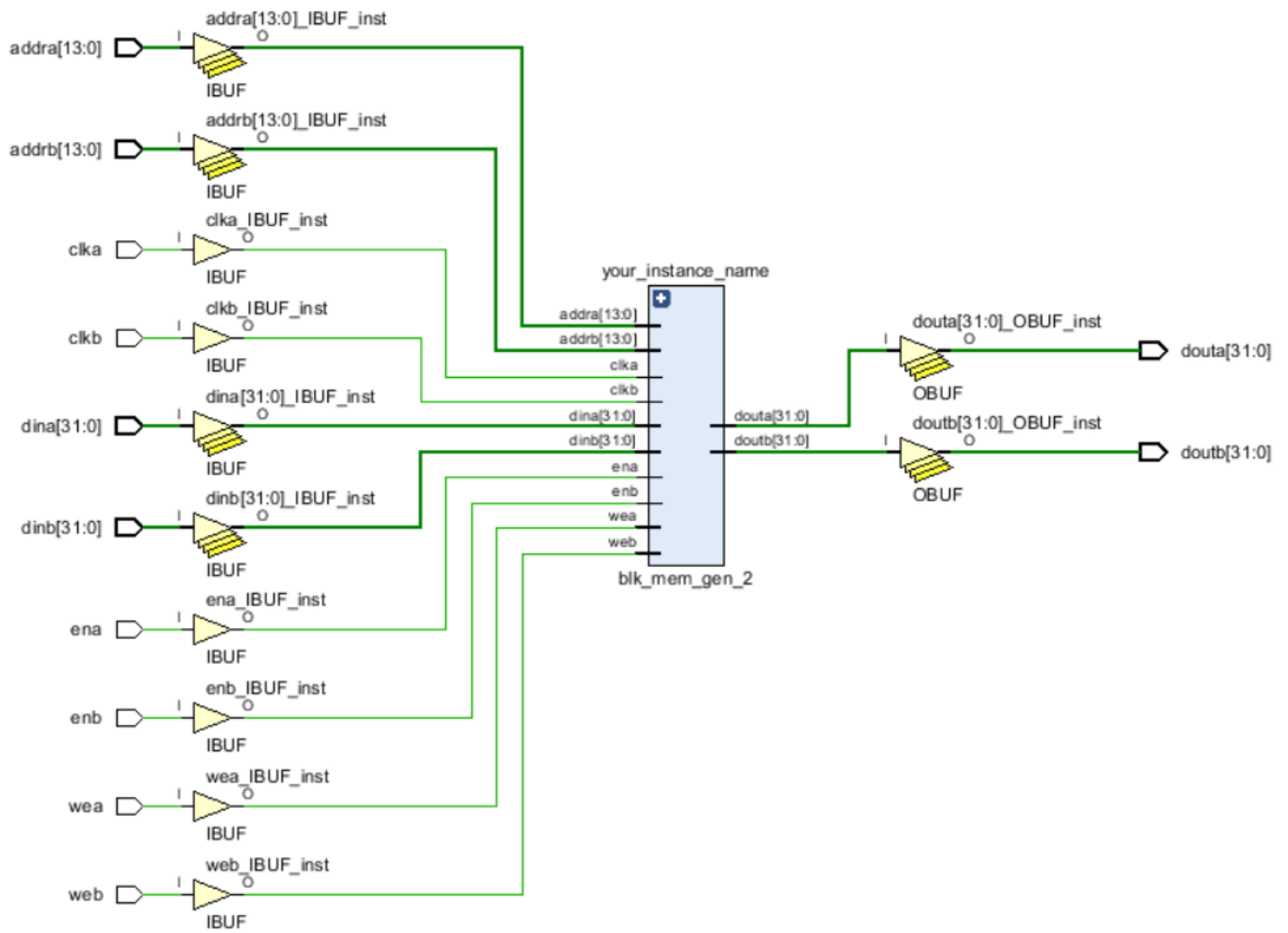
1. Slice Logic

Site Type	Used	Fixed	Available	Util%
Slice LUTs*	62	0	53200	0.12
LUT as Logic	62	0	53200	0.12
LUT as Memory	0	0	17400	0.00
Slice Registers	8	0	106400	<0.01
Register as Flip Flop	8	0	106400	<0.01
Register as Latch	0	0	106400	0.00
F7 Muxes	0	0	26600	0.00
F8 Muxes	0	0	13300	0.00

2. Memory

Site Type	Used	Fixed	Available	Util%
Block RAM Tile	14.5	0	140	10.36
RAMB36/FIFO*	14	0	140	10.00
RAMB36E1 only	14			
RAMB18	1	0	280	0.36
RAMB18E1 only	1			

RTL analysis schematic:



Distributed BRAM, size 256x32

//Verilog code [distributed_ram.v]

`timescale 1ns / 1ps

```
module distributed_ram (  
    input wire clk,  
    input wire we,  
    input wire [7:0] addr,  
    input wire [31:0] din,  
    output reg [31:0] dout);  
  
    (* ram_style = "distributed" *) reg [31:0] mem [0:255];  
  
    always @(posedge clk) begin  
        if (we)  
            mem[addr] <= din;  
        dout <= mem[addr];  
    end  
endmodule
```

1. Slice Logic

Site Type	Used	Fixed	Available	Util%
Slice LUTs*	128	0	53200	0.24
LUT as Logic	0	0	53200	0.00
LUT as Memory	128	0	17400	0.74
LUT as Distributed RAM	128	0		
LUT as Shift Register	0	0		
Slice Registers	32	0	106400	0.03
Register as Flip Flop	32	0	106400	0.03
Register as Latch	0	0	106400	0.00
F7 Muxes	64	0	26600	0.24
F8 Muxes	32	0	13300	0.24

RTL analysis schematic:

